Declaration of AI assistance

Journal of Number Theory

Author-prepared disclosure; not an official publisher form or signature.

Manuscript. The optimal log log q bound in the Pólya-Vinogradov inequality

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Claude through Claude Code assisted the earlier coauthored manuscript, formal development and checking programs. OpenAI Codex assisted the explicit 11/12 parameter conversion, proof and scope auditing, Mellin arguments, the general smooth Mobius equivalence, new Lean analytic modules, verification records, editing and typesetting. OpenAI's original zero-free theorem remains a credited external mathematical contribution whose proof sources are supplied in the repository.

Human oversight. The submitting author stated on 25 October 2026 that the text is a mix of human-written and AI-generated material and that all parts have been checked and understood by a human. This records that author declaration. Human authors retain responsibility for mathematical accuracy, citations and the final version; AI systems are not human coauthors.

Verification scope. The complete current source build, all 19 original and seven closed endpoint axiom reports, and all 20 analytic component reports passed. The reports use only propext, Classical.choice and Quot.sound. The separate written cyclotomic identification is not a new Lean statement, and the revised exposition is not a line-by-line formalisation. No independent zero-free proof is claimed.

Supporting record. The separate AI_PROVENANCE.md identifies the documented tasks and tools. No unprovided model version, human signature, confidential transcript or journal approval is asserted.

Policy reference. https://www.elsevier.com/about/policies-and-standards/generative-ai-policies-for-journals

Prepared on 25 October 2026. The author reviews this statement and completes any additional live-portal declarations before submission.